

TRANSPORT AND THE SUPERCRITICALITY GAP IN 3D NAVIER–STOKES, II: THE FLOW MAP

ADILKHAN SALKIMBAYEV

ABSTRACT. Part I showed that every term on the right-hand side of the strain equation of the three-dimensional Navier–Stokes equations fails, for a structural reason, to close an estimate at the critical level, and that advection contributes nothing to the budget of any functional of the Eulerian state. Here we move to the only object that records transport: the flow map, a curve in the group SDiff of volume-preserving diffeomorphisms. We give a dictionary between the Eulerian ledger of Part I and the geometry of this curve, and we make the resulting target precise in three statements. First, by the stochastic Lagrangian representation of Constantin and Iyer and the criterion of Beale, Kato and Majda, a smooth solution blows up exactly when the gradient of its stochastic flow map does; this quantity is scale invariant, and its proof uses transport of vorticity by the flow and recovery of velocity by Biot–Savart and nothing else. Second, the geometry that the energy controls cannot see deformation: by Shnirelman, the L^2 diameter of the volume-preserving diffeomorphism group of the three-dimensional cube is finite, while the right-invariant H^s distance controls the Lipschitz constant, and the diameter is infinite, for $s > 5/2$; the energy action sits at $s = 1$ and the critical action at $s = 3/2$. Third, an a priori Lagrangian bound does exist: the estimate of Crippa and De Lellis, transferred to the stochastic flow by a translation argument, gives Lipschitz control off small sets from the dissipation alone. It is scale invariant in form, but exactly as supercritical as the energy: combined with Leray’s lower bound on the blowup rate, it can exclude concentration of the flow map at Type I scale only under a dissipation bound that Leray’s argument already turns into a regularity criterion. Lagrangian bookkeeping by itself therefore gains nothing. What remains is a precise question: an estimate on the Lusin–Lipschitz profile of the stochastic flow that uses the Cauchy–Weber coupling of transport and Biot–Savart.

1. INTRODUCTION

We consider the incompressible Navier–Stokes equations on $\mathbb{T}^3 = (\mathbb{R}/2\pi\mathbb{Z})^3$,

$$\partial_t u - \nu \Delta u + (u \cdot \nabla)u + \nabla p = 0, \quad \nabla \cdot u = 0, \quad (1)$$

with smooth divergence-free data u_0 of mean zero, energy $E_0 = \frac{1}{2}\|u_0\|_{L^2}^2$, vorticity $\omega = \text{curl } u$, and maximal time $T^* \leq \infty$ of the smooth solution. The energy inequality gives

$$\frac{1}{2}\|u(t)\|_{L^2}^2 + \nu \int_0^t \|\nabla u\|_{L^2}^2 ds \leq E_0. \quad (2)$$

Part I [21] examined the strain equation $D_t S = -S^2 - \Omega^2 - \nabla^2 p + \nu \Delta S$ term by term. The local terms drive the singular configuration; the pressure contributes nothing to the L^2 strain budget and, in every dynamical state computed, works against the critical norm below L^2 ; the vorticity back-reaction saturates below the self-amplification; viscosity has supercritical scaling. The remaining term, advection, contributes identically zero to $\frac{d}{dt} \int F(\nabla u) dx$ for every F [21, Prop. 5.1]. If (1) is regular, the mechanism required by Miller’s model family [12, 20] must therefore act through transport, and it cannot be credited by any budget of the Eulerian state.

The object that records transport is the flow map X_t , a curve in the group $\text{SDiff}(\mathbb{T}^3)$ of volume-preserving diffeomorphisms. This paper does three things. Section 2 gives a dictionary between the ledger of Part I and the geometry of this curve, with the elementary facts that make it exact. Sections 3–5 prove or collect three statements that fix the target: a Lagrangian form of the regularity problem (Theorem 3.1), the blindness of the geometry controlled by the energy (Theorem 4.1 and Proposition 4.2), and the existence and exact supercriticality of an a priori Lagrangian bound (Theorem 5.2 and Proposition 5.4). Section 6 states the target as precise questions.

What is new, and what is not. The stochastic Lagrangian representation and the stochastic Cauchy formula are due to Constantin and Iyer [5]; the finite diameter of SDiff of the cube is due to Shnirelman [15, 16]; the quantitative Lagrangian estimate is due to Crippa and De Lellis [6]; the lower bound on the blowup rate is Leray’s [10]. Our contributions are the dictionary; Theorem 3.1, which is an immediate combination of [5] and [4], stated because it is the Lagrangian form of the problem and we have not found it written; the transfer of [6] to the stochastic flow (Lemma 5.3); the elementary Proposition 4.2; and Proposition 5.4, which measures the supercriticality of the Lagrangian bound against Leray’s lower bound. The questions of §6 are the point of the paper.

2. THE DICTIONARY

Let u be a smooth solution of (1) on $[0, T^*)$. The deterministic flow is $\dot{X}_t = u(X_t, t)$, $X_0 = \text{id}$; since $\nabla \cdot u = 0$, $X_t \in \text{SDiff}(\mathbb{T}^3)$. Its deformation gradient $F = \nabla_a X$ satisfies

$$\frac{d}{dt} F(t, a) = (\nabla u)(X_t(a), t) F(t, a), \quad \det F = 1. \quad (3)$$

The Eulerian velocity $u = \dot{X} \circ X^{-1}$ is unchanged if X is replaced by $X \circ h$ for a fixed $h \in \text{SDiff}$: Eulerian quantities are relabeling invariant (right invariant), and a functional of $u(t)$ carries no memory of how the fluid was carried to its present configuration. The deformation F is not relabeling invariant and carries exactly that memory.

The stochastic flow of Constantin and Iyer [5] is

$$dX_t = u(X_t, t) dt + \sqrt{2\nu} dW_t, \quad X_0 = \text{id}, \quad A_t = X_t^{-1}, \quad (4)$$

with W a three-dimensional Wiener process. Since the noise is spatially constant, ∇X again satisfies (3) pathwise, with ∇u evaluated along the noisy trajectory, and $X_t \in \text{SDiff}(\mathbb{T}^3)$ almost surely.

Theorem 2.1 (Constantin–Iyer [5]). *On the interval of existence of the smooth solution,*

$$u = \mathbb{E} \mathbb{P}[(\nabla^T A)(u_0 \circ A)], \quad \omega = \mathbb{E}[(\nabla X)\omega_0 \circ A], \quad (5)$$

where $\mathbb{P}$ is the Leray projection and $\mathbb{E}$ the expectation over W .

The first identity is the stochastic Weber formula; the second is the stochastic Cauchy formula. For $\nu = 0$ they reduce to the classical Weber and Cauchy formulas of the Euler equations.

Lemma 2.2 (material stretching). *Along any deterministic or stochastic trajectory and for any vector $\xi \neq 0$,*

$$\lambda_1 \leq \frac{d}{dt} \log |F\xi| = \hat{e} \cdot S \hat{e} \leq \lambda_3, \quad \hat{e} = \frac{F\xi}{|F\xi|},$$

where $\lambda_1 \leq \lambda_2 \leq \lambda_3$ are the eigenvalues of $S = \frac{1}{2}(\nabla u + \nabla u^T)$ at the current position. On the producing set $\{-\det S > 0\}$ one has $\lambda_3 \leq |\lambda_1|$, so $|\frac{d}{dt} \log |F\xi|| \leq |\lambda_1|$.

Proof. By (3), $\frac{d}{dt}|F\xi|^2 = 2F\xi \cdot (\nabla u)F\xi = 2F\xi \cdot SF\xi$. On the producing set $\lambda_2 > 0$, so $|\lambda_1| = \lambda_2 + \lambda_3 \geq \lambda_3$. $\square$

Corollary 2.3. *During a visit of a trajectory to the producing set, of compression $C = \int |\lambda_1| dt$ in the sense of [21, §6.5], $\log \|F\|$ and $\log \|F^{-1}\|$ change by at most C .*

The measurements of Part I are therefore measurements of the flow map: each producing event changes $\log \|\nabla X\|$ at the particle by a bounded amount, of order 1–4 in the simulations of [21, Table 4]. Table 1 collects the dictionary.

Part I (Eulerian)	flow map / SDiff	exact relation
velocity $u(t)$	right-trivialized tangent vector $\dot{X} \circ X^{-1}$	relabeling invariant
advection $u \cdot \nabla$	motion along the curve $t \mapsto X_t$	$\partial_t(f \circ X) = (D_t f) \circ X$
budget of $\int F(\nabla u) dx$	right-invariant function on TSDiff	advection contributes 0 [21, Prop. 5.1]
deformation (absent in Part I)	$F = \nabla X$, not right invariant	(3)
strain eigenvalues	stretching rates of material lines	Lemma 2.2
compression per visit C	bound on the change of $\log \ F\ $	Cor. 2.3
vorticity stretching	ω carried and stretched by F	(5), Cauchy
Biot–Savart	inverse inertia operator; recovers u from the transported ω	[1, 19]
pressure	second fundamental form of SDiff $\subset L^2(\mathbb{T}^3; \mathbb{T}^3)$; the projection $\mathbb{P}$ in (5)	[1, 13]
viscosity	Brownian perturbation of the paths, then averaging	(4), (5)
energy	L^2 speed of the deterministic curve	(2)
dissipation	right-invariant H^1 action	(2)

TABLE 1. The dictionary.

Two entries deserve comment. For the L^2 metric the inertia operator is the identity, the momentum is u , its curl is ω , and recovering u from ω is the Biot–Savart law; in the geometric language of [19], the transport of the curl of the momentum is the same for every inertia operator and the equation enters only through the constitutive law that recovers the velocity. Transport and Biot–Savart are thus the two halves of one loop: the flow carries and stretches the vorticity, and Biot–Savart turns the carried vorticity into the velocity that moves the flow. The pressure is the constraint force that keeps the curve inside SDiff; in (5) it is the projection $\mathbb{P}$.

3. THE LAGRANGIAN FORM OF THE PROBLEM

Theorem 3.1. *Let u be the smooth solution of (1) on its maximal interval $[0, T^*)$ and let X be its stochastic flow (4), and let $0 < T < \infty$. Then $T^* > T$ if and only if*

$$\sup_{t < \min(T, T^*)} \operatorname{ess\,sup}_W \|\nabla X_t\|_{L^\infty(\mathbb{T}^3)} < \infty.$$

In particular, if $T^ < \infty$ then ∇X_t is unbounded as $t \uparrow T^*$.*

Proof. If $T^* > T$, then u is smooth on $[0, T]$, so $\|\nabla u\|_{L^\infty}$ is bounded there, and (3) along every noisy trajectory gives $\|\nabla X_t\|_{L^\infty} \leq \exp \int_0^t \|\nabla u\|_{L^\infty} ds$ uniformly in W . Conversely, suppose $\|\nabla X_t\|_{L^\infty} \leq K$ for all $t < \min(T, T^*)$ and almost every W , and $T^* \leq T$. By (5), $|\omega(x, t)| \leq \mathbb{E}[|\nabla X_t|(A_t(x)) | \omega_0(A_t(x))|] \leq K \|\omega_0\|_{L^\infty}$ for $t < T^*$. Hence $\int_0^{T^*} \|\omega\|_{L^\infty} dt < \infty$, and by the Beale–Kato–Majda criterion, which holds for (1) [4, 9], the solution extends beyond T^* , a contradiction. $\square$

For global smooth solutions ∇X_t may of course grow without bound as $t \rightarrow \infty$; the theorem concerns finite intervals.

Remark 3.2. (i) Under the scaling $u_\lambda(x, t) = \lambda u(\lambda x, \lambda^2 t)$ (on $\mathbb{R}^3$, or with the period rescaled) the stochastic flow transforms as $X_\lambda(a, t) = \lambda^{-1} X(\lambda a, \lambda^2 t)$ with Brownian rescaling of W , and ∇X is invariant: Theorem 3.1 is a critical criterion.

(ii) The proof uses two facts: the vorticity is carried and stretched by the flow (the Cauchy formula), and the velocity is recovered from the vorticity with only a logarithmic loss (the Biot–Savart estimate inside [4]). These are the two halves of the loop in §2. Neither Tao’s averaged equation [17], whose nonlinearity is not in general the coadjoint action and admits no circulation theorem, nor Miller’s strain models [11, 12], which have no advection, possess both.

(iii) For the deterministic flow only the forward implication holds: without noise there is no Cauchy formula for (1). The stochastic flow is the correct Lagrangian object for the viscous equations.

4. THE GEOMETRY CONTROLLED BY THE ENERGY IS BLIND

The deterministic curve $t \mapsto X_t$ has L^2 speed $\|u(t)\|_{L^2} \leq \sqrt{2E_0}$ and, by (2), right-invariant H^1 action $\int_0^t \|\nabla u\|_{L^2}^2 ds \leq E_0/\nu$. These are the only unconditional facts about its geometry.

Theorem 4.1 (Shnirelman [15, 16]). *Let K^3 be the unit cube. The diameter of $\text{SDiff}(K^3)$ in the geodesic distance of the right-invariant L^2 metric is finite. The completion of $\text{SDiff}(K^3)$ in this distance is the set of all measure-preserving maps of K^3 .*

In two dimensions the diameter is infinite [8]. The diameter statement extends to contractible manifolds of dimension at least three [2]; for $\mathbb{T}^3$, which is not contractible, we do not claim it. Under the dilation $x \mapsto rx$ a path of volume-preserving maps of the unit cube becomes a path of maps of the cube of side r whose L^2 length is multiplied by $r^{5/2}$. So, within the group of the cube (transferring this to $\mathbb{T}^3$ would need Shnirelman’s construction for compactly supported maps, which we have not checked), a deformation of arbitrary complexity localized in a cube of side r costs L^2 length at most $Dr^{5/2}$, where D is the diameter of Theorem 4.1; over a time interval of length τ the energy alone allows length $\tau\sqrt{2E_0}$, which exceeds $D(\nu\tau)^{5/4}$, the cost at the parabolic scale $r = (\nu\tau)^{1/2}$, for all small τ . The length budget of the energy does not exclude any localized deformation, however folded.

At the other end of the Sobolev scale the distance does see deformation.

Proposition 4.2. *Let $s > 5/2$ and let d_s be the geodesic distance of the right-invariant H^s metric on $\text{SDiff}(\mathbb{T}^3)$. There is $c_s > 0$ such that $d_s(\text{id}, \varphi) \geq c_s \log \text{Lip}(\varphi)$ for every φ . In particular $\text{SDiff}(\mathbb{T}^3)$ has infinite diameter in d_s .*

Proof. For a path φ_t with $\dot{\varphi}_t = v_t \circ \varphi_t$, (3) gives $\log \text{Lip}(\varphi_1) \leq \int_0^1 \|\nabla v_t\|_{L^\infty} dt \leq C \int_0^1 \|v_t\|_{H^s} dt$ by Sobolev embedding. The shear maps $\psi_a(x, y, z) = (x + a \sin y, y, z)$ lie in $\text{SDiff}(\mathbb{T}^3)$, are joined to the identity by $t \mapsto \psi_{ta}$, and have $\text{Lip}(\psi_a) \geq a$. $\square$

The exponents line up with the scaling of (1). The H^s length $\int \|u\|_{H^s} dt$ is critical at $s = 5/2$ and the H^s action $\int \|u\|_{H^s}^2 dt$ is critical at $s = 3/2$; the energy controls the action at $s = 1$. The distance controls the Lipschitz constant exactly from the critical length exponent on. Whether $\text{SDiff}(K^3)$ has finite or infinite diameter for the H^s metrics with $0 < s \leq 5/2$ is, to our knowledge, open; Bauer and Maor list the corresponding question for volume-preserving maps among their open problems [2]. If the H^1 diameter is finite, the dissipation-level geometry is blind in the same sense as the energy-level geometry of Theorem 4.1. Either way, the target of this paper cannot be reached by bounding the size of the curve in a metric that the a priori information controls.

5. AN A PRIORI LAGRANGIAN BOUND, AND ITS SUPERCRITICALITY

Definition 5.1. For a map $Y : \mathbb{T}^3 \rightarrow \mathbb{T}^3$ and $\varepsilon > 0$, the *Lusin–Lipschitz profile* of Y is $L_Y(\varepsilon) = \inf\{\text{Lip}(Y|_K) : K \subset \mathbb{T}^3 \text{ measurable, } |\mathbb{T}^3 \setminus K| \leq \varepsilon\}$. It is nonincreasing, and for continuous Y , $\lim_{\varepsilon \rightarrow 0} L_Y(\varepsilon) = \text{Lip}(Y)$.

By Theorem 3.1, regularity up to T^* is equivalent to a bound on $L_{X_t}(0^+)$ uniform in $t < T^*$ and in the noise. What does the a priori information give?

Theorem 5.2 (after Crippa–De Lellis [6]). *There is an absolute constant C such that for every smooth divergence-free field v on $\mathbb{T}^3 \times [s, t]$ with flow Y from time s to time t , and every $\varepsilon > 0$,*

$$\log L_Y(\varepsilon) \leq C \left(1 + \varepsilon^{-1/2} \int_s^t \|\nabla v(r)\|_{L^2} dr \right).$$

Sketch, following [6]. Let M denote the maximal function truncated at a fixed radius R_0 , and for $r \leq R_0$ let $Q_r(t, x) = \frac{1}{|B_r|} \int_{B_r(x)} \log(1 + |Y_t(x) - Y_t(y)|/r) dy$, with distances computed on lifts. By the pointwise inequality $|v(a) - v(b)| \leq C|a - b|(M|\nabla v|(a) + M|\nabla v|(b))$,

$$\partial_t Q_r(t, x) \leq \frac{C}{|B_r|} \int_{B_r(x)} (g_t(Y_t(x)) + g_t(Y_t(y))) dy, \quad g_t = M|\nabla v(t)|.$$

Taking the supremum over r ,

$$Q^*(t, x) = \sup_r Q_r(t, x) \leq \log 2 + CG(x), \quad G(x) = \int_s^t (g_\sigma(Y_\sigma(x)) + M(g_\sigma \circ Y_\sigma)(x)) d\sigma,$$

the constant $\log 2$ being the value at the initial time. Since Y_σ preserves measure and M is bounded on L^2 , $\|G\|_{L^2} \leq C \int_s^t \|\nabla v\|_{L^2}$. For x, y at distance r , averaging the triangle inequality over a ball of radius $r/2$ contained in $B_{2r}(x) \cap B_{2r}(y)$ and using $\log(1+a+b) \leq \log(1+a) + \log(1+b)$ gives $\log(1 + |Y(x) - Y(y)|/r) \leq C(1 + Q^*(x) + Q^*(y))$; pairs at distance above $R_0/2$ are handled by the boundedness of $\mathbb{T}^3$. So Y is $e^{C(1+m)}$ -Lipschitz on $\{G \leq m\}$, whose complement has measure at most $\|G\|_{L^2}^2/m^2$. Choosing $m = \varepsilon^{-1/2}\|G\|_{L^2}$ proves the claim. $\square$

Lemma 5.3. *Theorem 5.2 holds for every realization of the stochastic flow (4), with $v = u$.*

Proof. $Y_t = X_t - \sqrt{2\nu}W_t$ is the flow of the field $v^W(y, t) = u(y + \sqrt{2\nu}W_t, t)$, which is divergence free with $\|\nabla v^W(t)\|_{L^2} = \|\nabla u(t)\|_{L^2}$, and X_t, Y_t differ by a constant translation, so they have the same Lusin–Lipschitz profile. $\square$

By (2), $\int_s^t \|\nabla u\|_{L^2} \leq (t-s)^{1/2}(E_0/\nu)^{1/2}$. The flow map of every Leray–Hopf-bounded solution is therefore Lipschitz, with a constant controlled by the energy, off sets of any prescribed small measure. The estimate is scale invariant in form: $\varepsilon^{1/2}$ and $\int \|\nabla u\|_{L^2} dt$ both have the dimension

of a volume to the power $1/2$. Its supercriticality lies in the input, and it can be measured against Leray's lower bound on the blowup rate: if $T^* < \infty$ then [10, 14]

$$\|\nabla u(t)\|_{L^2} \geq c_L \nu^{3/4} (T^* - t)^{-1/4} \quad (t < T^*). \quad (6)$$

Write $D(\tau) = \nu \int_{T^*-\tau}^{T^*} \|\nabla u\|_{L^2}^2 dt$ for the dissipation in the final interval of length τ ; by (2), $D(\tau) \rightarrow 0$, and by (6), $D(\tau) \geq 2c_L^2 \nu^{5/2} \tau^{1/2}$.

For a flow Y from time s to time t and $q > 0$, let $\mu_q(Y) = |\{x : Q^*(x) > q\}|$ be the measure of the set on which Y has stretched nearby pairs by more than a factor of order e^q , in the averaged sense of Q^* .

Proposition 5.4. *Suppose $T^* < \infty$, and suppose the flow concentrates at Type I scale in the following sense: for some $q_0 > 2 \log 2$, some $c_0 > 0$ and all small τ , there are $t < T^*$ and a realization of the noise for which the stochastic flow Y from $T^* - \tau$ to t has $\mu_{q_0}(Y) \geq c_0(\nu\tau)^{3/2}$. Then*

$$D(\tau) \geq \kappa_* \nu^{5/2} \tau^{1/2}, \quad \kappa_* = c_0 q_0^2 / (4C^4).$$

In particular, an a priori bound $D(\tau) \leq \kappa \nu^{5/2} \tau^{1/2}$ with $\kappa < \kappa_$ would exclude such concentration, while the same bound with $\kappa < 2c_L^2$ excludes blowup outright by (6).*

Proof. On $\{Q^* > q_0\}$ we have $CG > q_0 - \log 2 > q_0/2$. By the proof of Theorem 5.2 and Lemma 5.3, $c_0(\nu\tau)^{3/2} q_0^2 / (4C^2) \leq \|G\|_{L^2}^2 \leq C^2 (\int_{T^*-\tau}^t \|\nabla u\|_{L^2}^2) \leq C^2 \tau D(\tau) / \nu$. $\square$

The a priori information gives $D(\tau) = o(1)$; what Proposition 5.4 needs is $D(\tau) = O(\tau^{1/2})$ with a small constant. The missing factor $\tau^{1/2}$ is one power of the parabolic length $(\nu\tau)^{1/2}$, which is the gap between the energy, of dimension L at fixed ν , and a critical quantity, of dimension L^0 . The Lagrangian estimate is exactly as supercritical as the energy it is built from, and Leray's classical argument already extracts a regularity criterion of the same form from the same kind of bound. Lagrangian bookkeeping of this kind gains nothing on its own.

6. THE TARGET

The three statements fix what the missing estimate must be.

- (1) It must control the stochastic flow map: by Theorem 3.1 this is equivalent to regularity, at the critical level.
- (2) It cannot come from the size of the curve $t \mapsto X_t$ in any metric the a priori information controls: the energy metric is blind (Theorem 4.1), and the metric that sees Lipschitz constants is critical in length (Proposition 4.2).
- (3) It cannot come from Lagrangian bookkeeping driven by $\int \|\nabla u\|$ alone (Proposition 5.4). It must use the Cauchy–Weber coupling (5): the flow carries and stretches the vorticity, and Biot–Savart turns the carried vorticity into the velocity that drives the flow.

Question 6.1 (critical concentration). *Is there a function $\kappa(q) \rightarrow 0$ as $q \rightarrow \infty$, depending only on scale-invariant quantities of the data, such that for every smooth solution of (1), every interval $[T - \tau, T]$ inside its interval of existence, and almost every realization of the noise, the stochastic flow Y from $T - \tau$ to T satisfies $\mu_q(Y) \leq \kappa(q)(\nu\tau)^{3/2}$? A positive answer excludes concentration of the flow at Type I scale in the sense of Proposition 5.4. The a priori information gives only $\mu_q(Y) \leq 4C^4 \tau D / (\nu q^2)$, with D the dissipation on the interval, which is $o(\tau)$ and not $O(\tau^{3/2})$.*

Question 6.2 (full control). *Is $\text{ess sup}_W \|Q^*\|_{L^\infty}$, for the stochastic flow from 0 to t , bounded for t in bounded intervals in terms of the data? Since $Q^* \leq \log(1 + \text{Lip } Y)$ and, by the proof of Theorem 5.2, $\text{Lip } Y \leq e^{C(1+2\|Q^*\|_{L^\infty})}$, this is equivalent to regularity by Theorem 3.1.*

Question 6.1 is weaker than regularity and stronger than anything the energy gives. In the language of Part I it asks for the escalation of producing events to be controlled at the parabolic scale. It bounds the measure of the set of *labels* that have been stretched; Part IV [23, Rem. 2.8] shows that this is a different quantity from the current extent of the structures that do the stretching, which is what a viscous threshold must control. Material lines stretch wherever $\lambda_3 > 0$, but the self-amplification of strain happens on the producing set, and by Corollary 2.3 each visit to that set changes $\log \|\nabla X\|$ by a bounded amount, observed to be of order one. The concentration that Question 6.1 rules out would therefore, in the picture of Part I, need many such visits whose contributions accumulate on a common, small set of trajectories; stretching outside the producing set is not self-amplifying and is controlled by the same strain. A mechanism that decorrelates successive visits would be the physical content of a positive answer; the registered hand-off measurements of Part III [22] test it in simulations, and Part IV [23] makes the events rigorous. The Eulerian criterion of Yoneda, Goto and Tsuruhashi [18], which asks the normalized efficiency of scale-to-scale vortex stretching to grow sublinearly in the wavenumber, is the frequency-space relative of the same idea.

Remark 6.3. A positive answer to Question 6.2 is regularity; a positive answer to Question 6.1 excludes Type I concentration of the flow. A negative answer for a model equation is informative only if the model has the Cauchy–Weber structure: the equivalence of Theorem 3.1 is not available without it, and the known blowup constructions for models [17, 11] lack it.

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ALMATY, KAZAKHSTAN

Email address: adilsalkimbayev@gmail.com